MLOPS Assignment-2

Tracking Model Performance Using MLflow

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Problem statement:

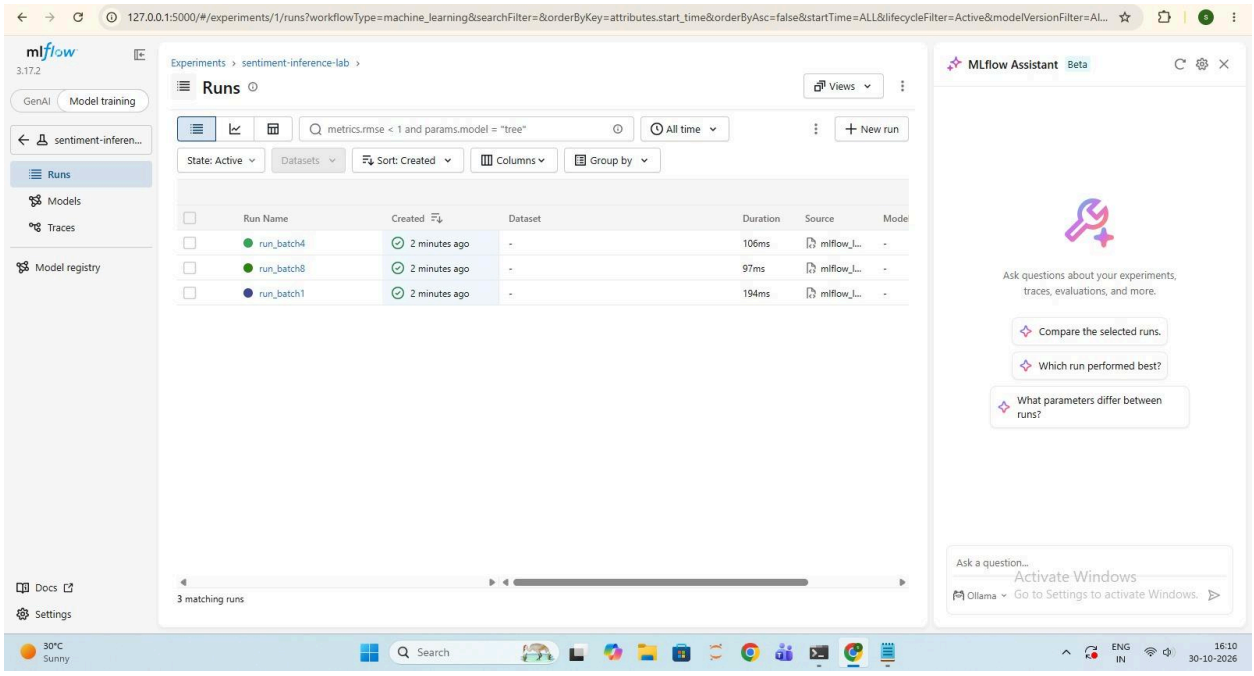
Tracking the comparing machine learning experiments manually makes it difficult to monitor model performance and maintain experiment records. A systematic approach is needed to record parameters, metrics and artifacts for easy comparison.

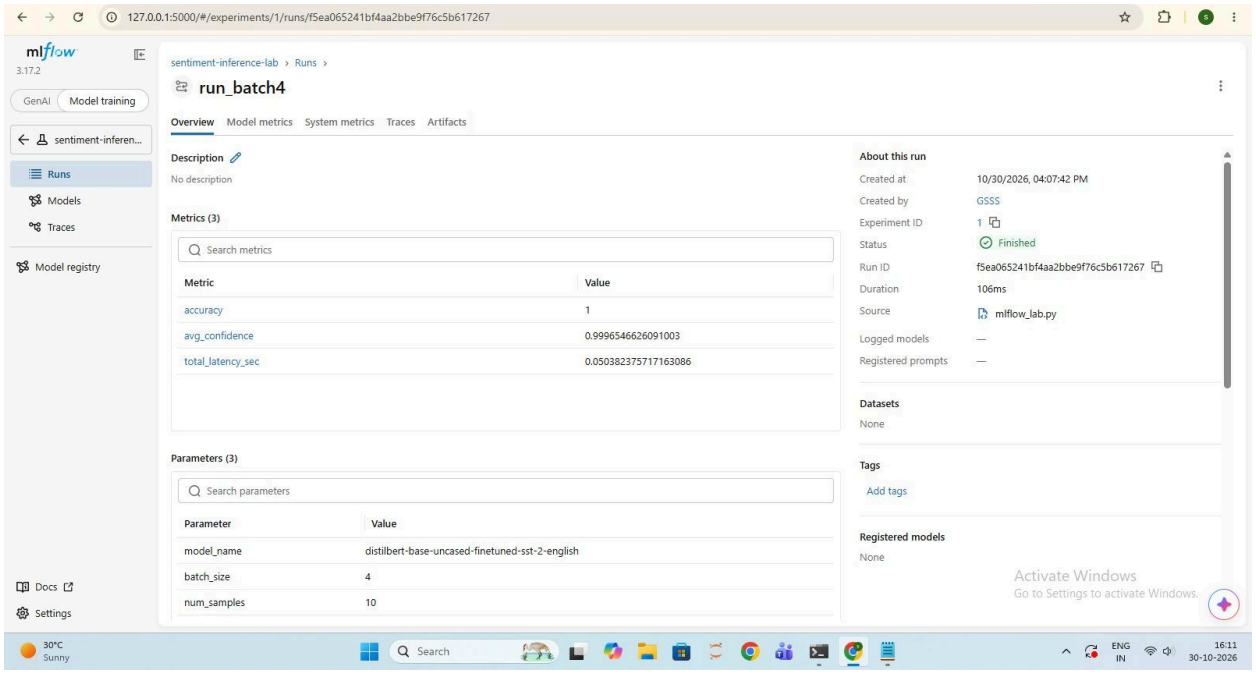
Objective:

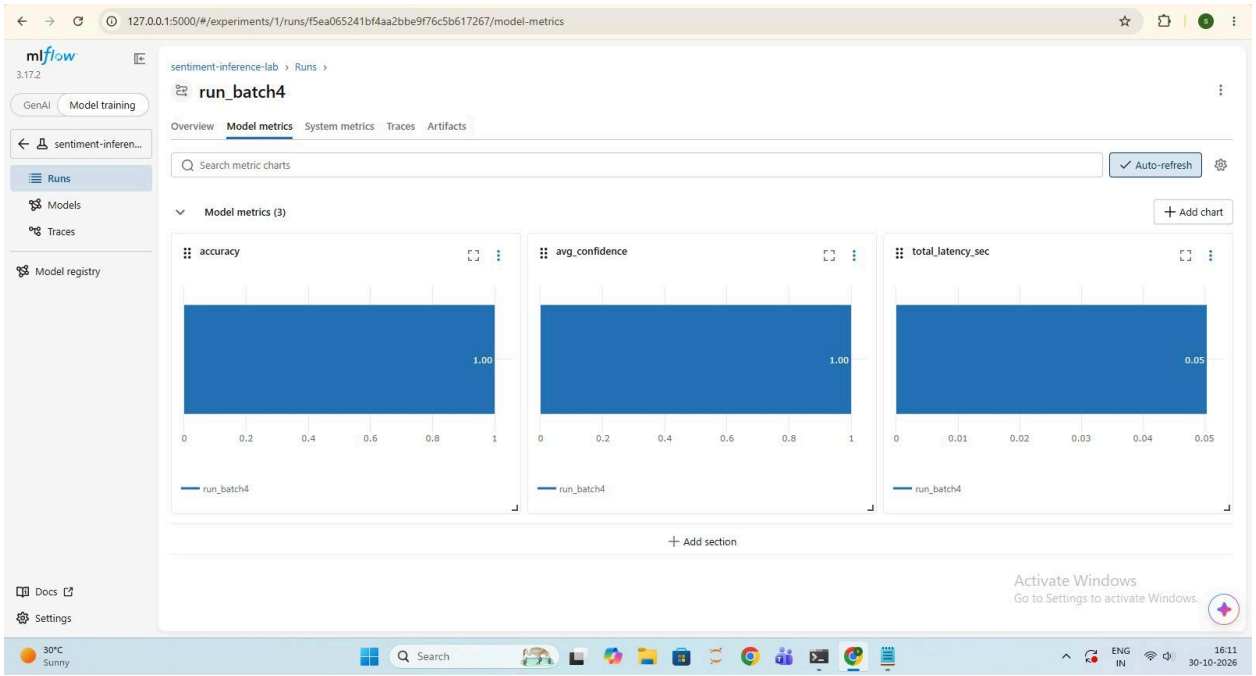
To track the parameters, metrics and artifacts of a pre-defined sentiment-analysis model using ML flow and compare multiple experiment runs.

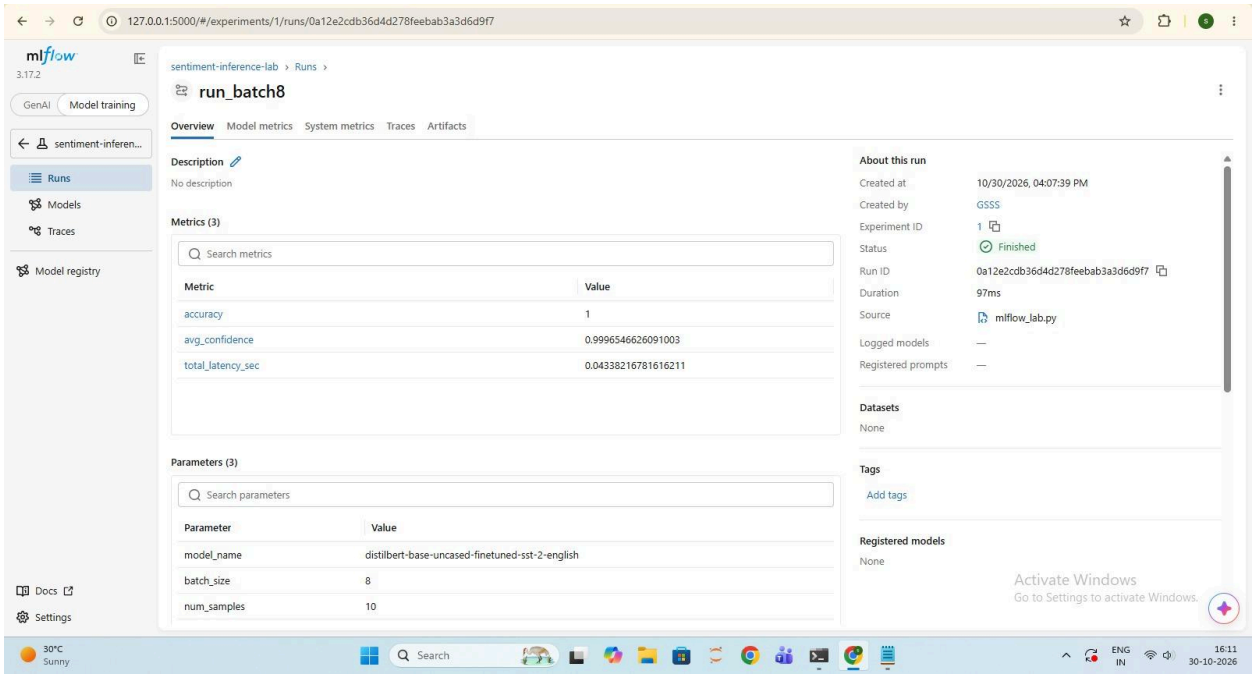
Result:

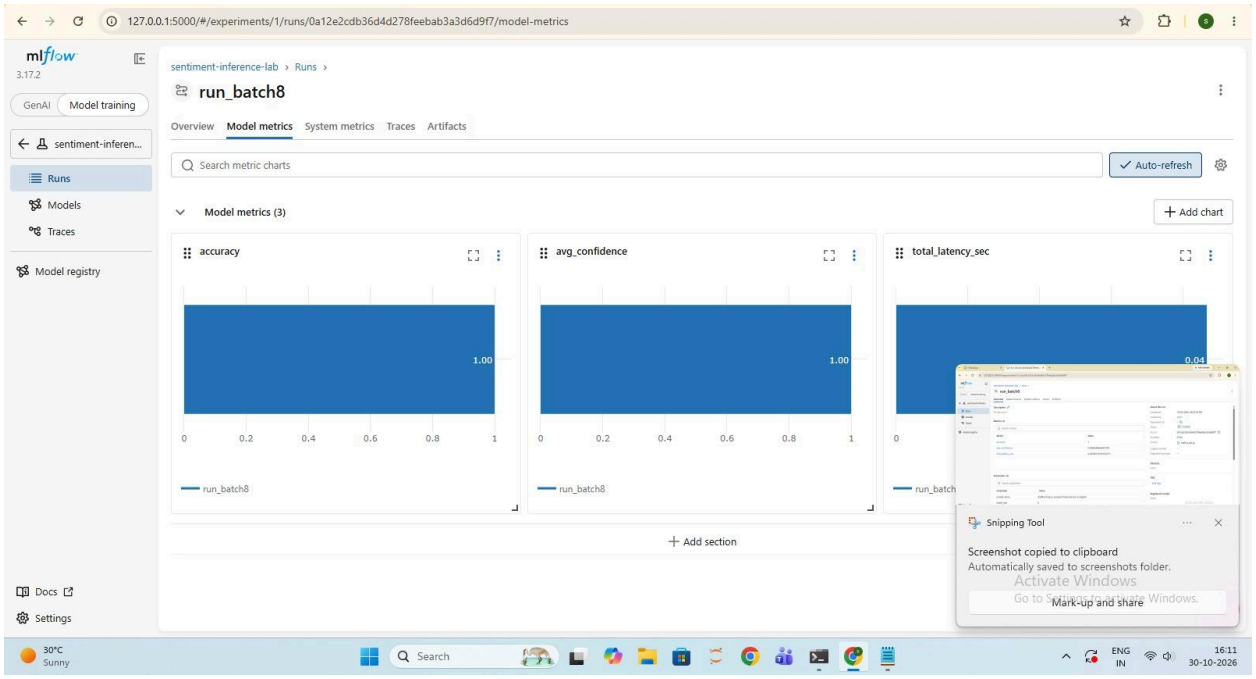
MLflow run table

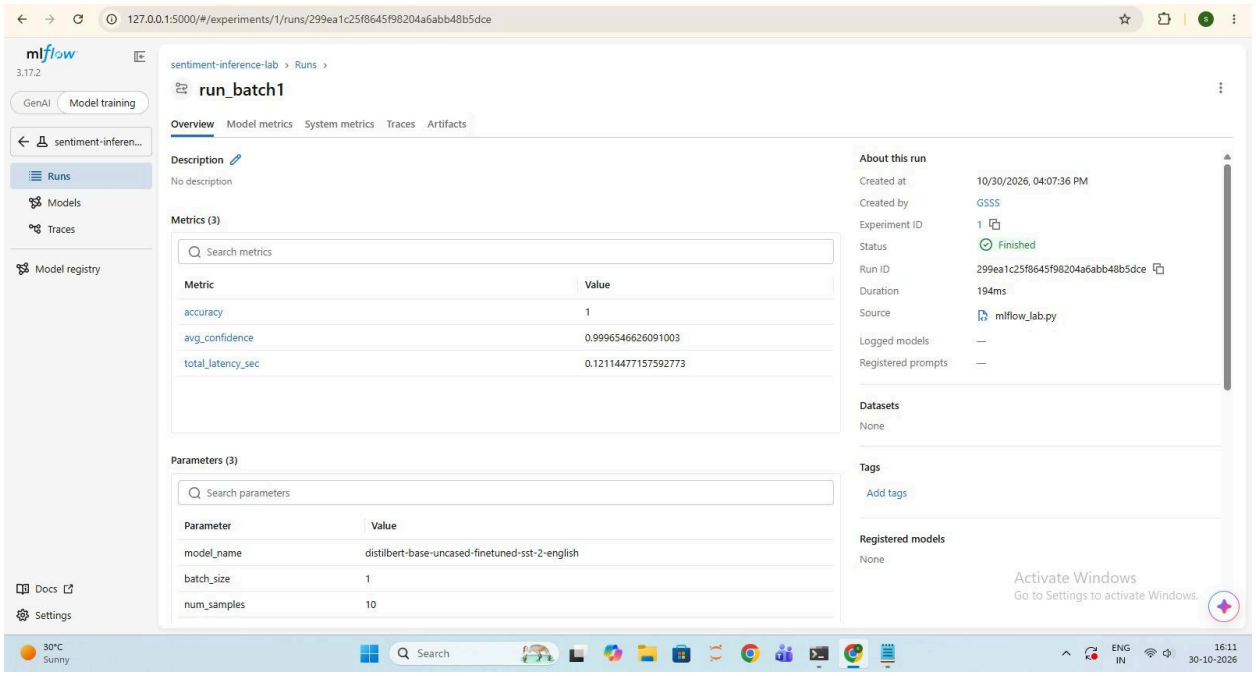


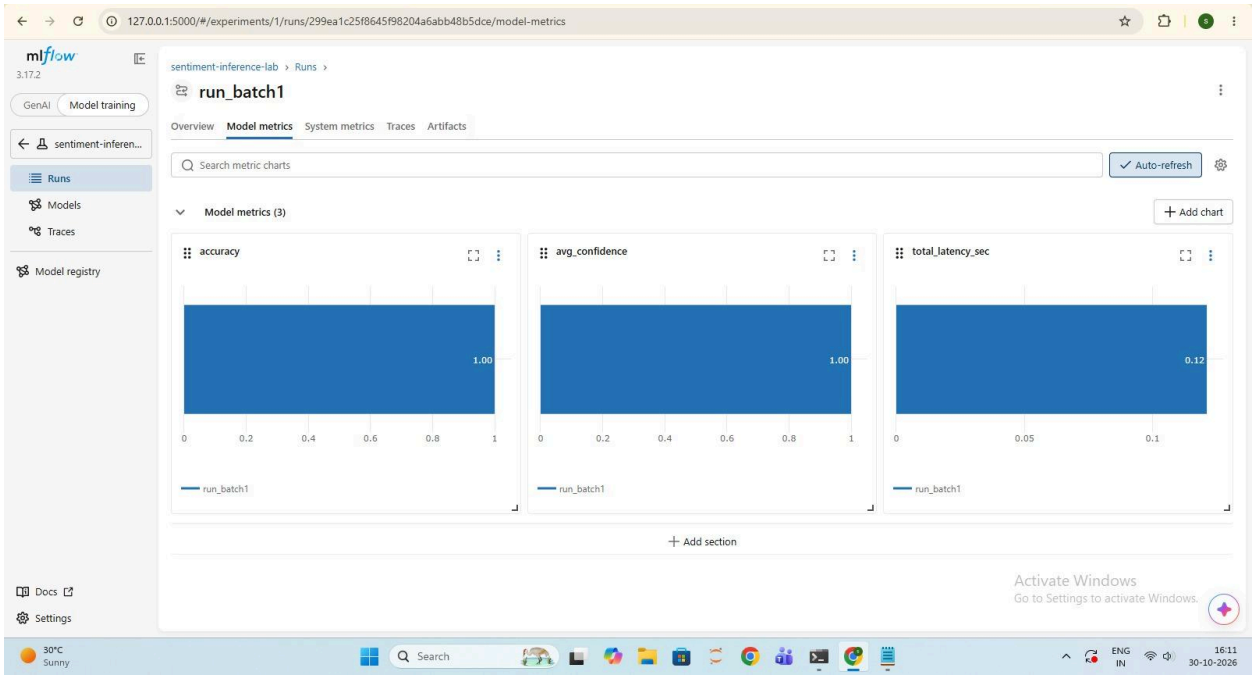

127.0.0.1:5000/#/experiments/1/runs/f5ea065241bf4aa2bbe9f76c5b617267
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch4
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (3)
Search metrics
Metric
Value
accuracy
1
avg_confidence
0.9996546626091003
total_latency_sec
0.050382375717163086
Parameters (3)
Search parameters
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
4
num_samples
10
About this run
Created at
10/30/2026, 04:07:42 PM
Created by
GSSS
Experiment ID
1
Status
Finished
Run ID
f5ea065241bf4aa2bbe9f76c5b617267
Duration
106ms
Source
mlflow_lab.py
Logged models
—
Registered prompts
—
Datasets
None
Tags
Add tags
Registered models
None
Activate Windows
Go to Settings to activate Windows.
30°C
Sunny
Search
ENG
IN
16:11
30-10-2026


127.0.0.1:5000/#/experiments/1/runs/f5ea065241bf4aa2bbe9f76c5b617267/model-metrics
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch4
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
1.00
0.05
0
0.2
0.4
0.6
0.8
1
0
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0.02
0.03
0.04
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run_batch4
Add section
Activate Windows
Go to Settings to activate Windows.
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30-10-2026

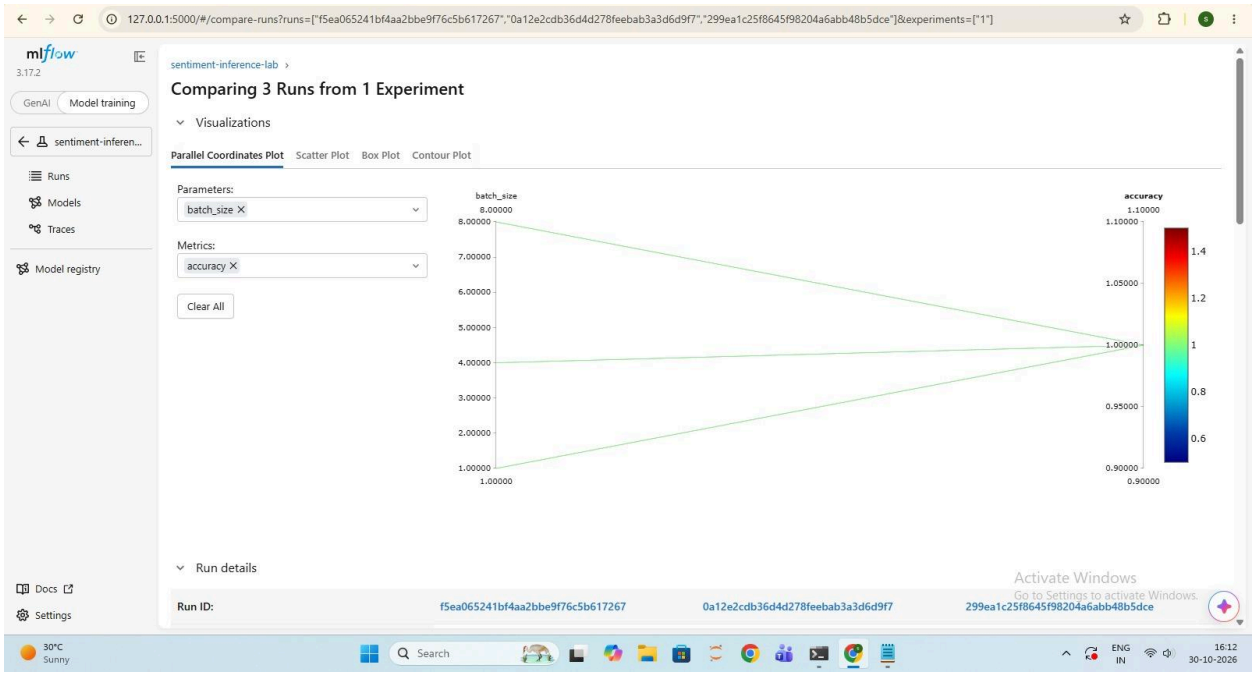

127.0.0.1:5000/#/experiments/1/runs/0a12e2cdb36d4d278feebab3a3d6d9f7
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (3)
Search metrics
Metric
Value
accuracy
1
avg_confidence
0.9966546626091003
total_latency_sec
0.04338216781616211
Parameters (3)
Search parameters
Parameter
Value
model_name
distilbert-base-uncased-finetuned-sst-2-english
batch_size
8
num_samples
10
About this run
Created at
10/30/2026, 04:07:39 PM
Created by
GSSS
Experiment ID
1
Status
Finished
Run ID
0a12e2cdb36d4d278feebab3a3d6d9f7
Duration
97ms
Source
mlflow_lab.py
Logged models
—
Registered prompts
—
Datasets
None
Tags
Add tags
Registered models
None
Activate Windows
Go to Settings to activate Windows.
30°C
Sunny
Search
ENG
IN
16:11
30-10-2026


127.0.0.1:5000/#/experiments/1/runs/0a12e2cdb36d4d278feebab3a3d6d9f7/model-metrics
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
1.00
0.04
0
0.2
0.4
0.6
0.8
1
run_batch8
Add section
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share
Activate Windows
Go to Settings to activate Windows.
30°C
Sunny
Search
ENG
IN
16:11
30-10-2026





MLflow compare view



cmd:

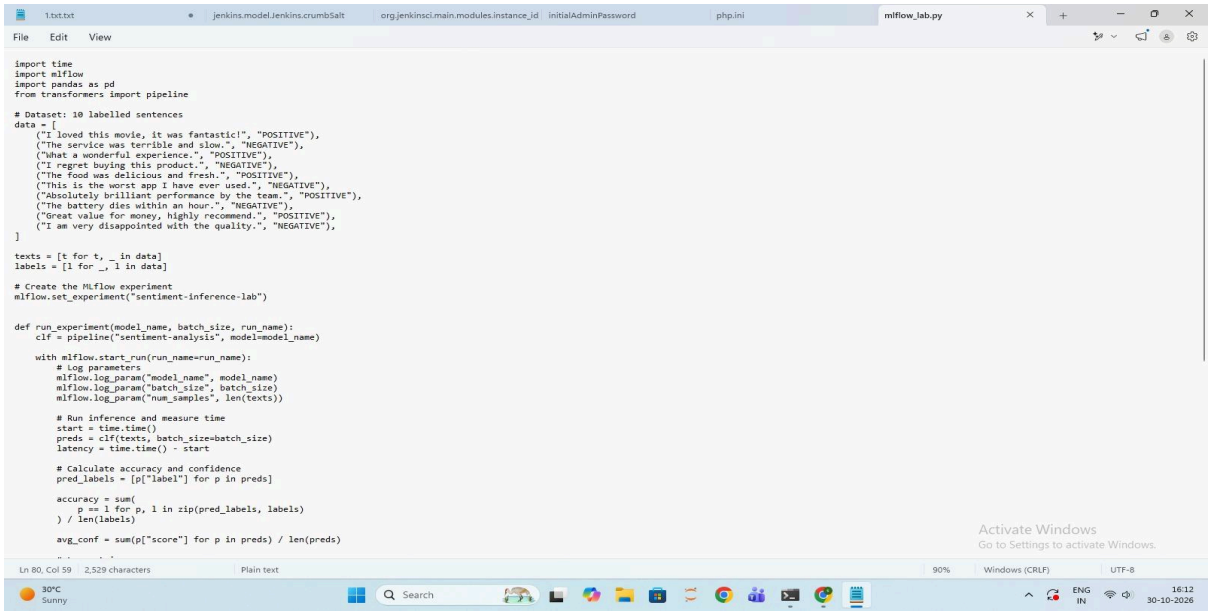
```
Command Prompt
vironment variable. For more details, see https://huggingface.co/docs/huggingface_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: https:
//docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development
  warnings.warn(message)
C:\Users\GSSS\Desktop\MLOps_Lab2\mlops-env\Lib\site-packages\huggingface_hub\file_download.py:149: UserWarning: `huggingface_hub` cache-system uses symlinks by default to eff
iciently store duplicated files but your machine does not support them in C:\Users\GSSS\.cache\huggingface\hub. Caching files will still work but in a degraded version that m
ight require more space on your disk. This warning can be disabled by setting the `HF_HUB_DISABLE_SYMLINKS_WARNING` environment variable. For more details, see https://huggin
gface.co/docs/huggingface_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: https:
//docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development
  warnings.warn(message)
model.safetensors: downloading bytes: |  250MB, 4.22MB/s
model.safetensors: reconstructing file: 100%|| 268MB /  268MB, 16.3MB/s
Loading weights: 100%|| 104/104 [00:00<00:00, 1164.51it/s]
tokenizer_config.json: 100%|| 48.0/48.0 [00:00<?, ?B/s]
vocab.txt: 100%|| 232k/232k [00:00<00:00, 762kB/s]
Model downloaded successfully!
```

```
(mlops-env) C:\Users\GSSS\Desktop\MLOps_Lab2>python mlflow_lab.py
2026/10/30 16:07:32 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2026/10/30 16:07:32 INFO mlflow.store.db.utils: Updating database tables
2026/10/30 16:07:33 INFO mlflow.tracking.fluent: Experiment with name 'sentiment-inference-lab' does not exist. Creating a new experiment.
Loading weights: 100%| 104/104 [00:00<00:00, 1188.07it/s]
run_batch1 -> accuracy: 1.0 latency: 0.121
Loading weights: 100%| 104/104 [00:00<00:00, 1157.43it/s]
run_batch8 -> accuracy: 1.0 latency: 0.043
Loading weights: 100%| 104/104 [00:00<00:00, 965.89it/s]
run_batch4 -> accuracy: 1.0 latency: 0.05

(mlops-env) C:\Users\GSSS\Desktop\MLOps_Lab2>
```

Activate Windows
Go to Settings to activate Windows.

code:



```
import time
import mlflow
import pandas as pd
from transformers import pipeline

# Dataset: 10 labelled sentences
data = [
    ("I loved this movie, it was fantastic!", "POSITIVE"),
    ("The service was terrible and slow.", "NEGATIVE"),
    ("What a wonderful experience.", "POSITIVE"),
    ("I regret buying this product.", "NEGATIVE"),
    ("The food was delicious and fresh.", "POSITIVE"),
    ("This is the worst app I have ever used.", "NEGATIVE"),
    ("Absolutely brilliant performance by the team.", "POSITIVE"),
    ("The battery dies within an hour.", "NEGATIVE"),
    ("Great value for money, highly recommend.", "POSITIVE"),
    ("I am very disappointed with the quality.", "NEGATIVE"),
]

texts = [t for t, _ in data]
labels = [l for _, l in data]

# Create the MLflow experiment
mlflow.set_experiment("sentiment-inference-lab")


def run_experiment(model_name, batch_size, run_name):
    clf = pipeline("sentiment-analysis", model=model_name)

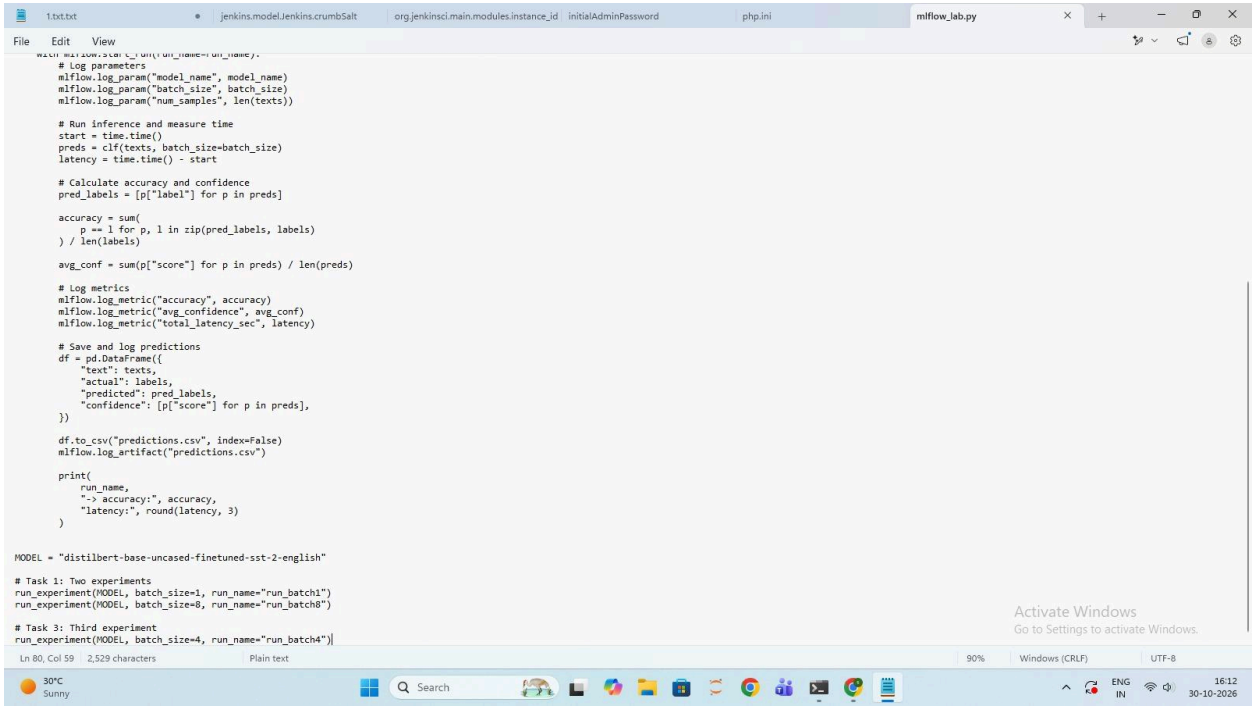
    with mlflow.start_run(run_name=run_name):
        # Log parameters
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        # Run inference and measure time
        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start

        # Calculate accuracy and confidence
        pred_labels = [p["label"] for p in preds]

        accuracy = sum(
            p == l for p, l in zip(pred_labels, labels)
        ) / len(labels)

        avg_conf = sum(p["score"] for p in preds) / len(preds)
```



```
        # Log parameters
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        # Run inference and measure time
        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start

        # Calculate accuracy and confidence
        pred_labels = [p["label"] for p in preds]

        accuracy = sum(
            p == l for p, l in zip(pred_labels, labels)
        ) / len(labels)

        avg_conf = sum(p["score"] for p in preds) / len(preds)

        # Log metrics
        mlflow.log_metric("accuracy", accuracy)
        mlflow.log_metric("avg_confidence", avg_conf)
        mlflow.log_metric("total_latency_sec", latency)

        # Save and log predictions
        df = pd.DataFrame({
            "text": texts,
            "actual": labels,
            "predicted": pred_labels,
            "confidence": [p["score"] for p in preds],
        })

        df.to_csv("predictions.csv", index=False)
        mlflow.log_artifact("predictions.csv")

        print(
            run_name,
            "-> accuracy:", accuracy,
            "latency:", round(latency, 3)
        )


MODEL = "distilbert-base-uncased-finetuned-sst-2-english"

# Task 1: Two experiments
run_experiment(MODEL, batch_size=1, run_name="run_batch1")
run_experiment(MODEL, batch_size=8, run_name="run_batch8")

# Task 3: Third experiment
run_experiment(MODEL, batch_size=4, run_name="run_batch4")
```

Question and answers:

1. Which run had the lowest latency, and why might that be?

run_batch8 recorded the lowest latency of 0.043 seconds. Processing multiple sentences in a batch can reduce overhead and make inference faster.

2. Did accuracy change between runs? Why or why not?

Accuracy remained the same across all three runs. All runs achieved an accuracy of 1.0 (100%) because the same pre-trained sentiment analysis model was used on the same test data.

3. Name two benefits of using MLflow here instead of printing results in the terminal

- MLflow records experiment parameters, performance metrics, and output files automatically.
- Its user interface makes it easier to compare results across multiple experiment runs.